

PROBLEM STATEMENT ID	THEMES	PROBLEM STATEMENT	TYPE(SOFTWARE/HARDWARE)	DESCRIPTION
1	SMART AUTOMATION	Fusion 360-based industrial component design and NC code generation.	HARDWARE	This problem statement focuses on developing a CAD/CAM-based manufacturing solution for an industrial component using Autodesk Fusion 360. Students must design the component, define appropriate machine and material details, select suitable tools from the tool library, and generate optimized toolpaths and NC/G-code. The generated code should be simulated and verified for machining accuracy and efficiency. The project should solve a practical industrial problem while demonstrating skills in CAD/CAM, CNC programming, toolpath optimization, and machining fundamentals
2	SMART AUTOMATION	Real-time monitoring of online certificate issuance.	SOFTWARE	The issuance of Caste and other certificates by Revenue Department need real-time monitoring to evaluate the resource allocation and demand for such certificates. The allocation of resources at present is done without any analysis leading to huge backlogs in some Sub-division where the application load is very high. Any effective monitoring at District and Central level with detailed evaluation shall enable providing better allocation of resources for issuanc of such certificates
3	FITNESS & SPORTS	AI-based mobile platform for sports talent assessment.	SOFTWARE	AI-Powered Mobile Platform for Democratizing Sports Talent Assessment is a mobile application that uses AI, Computer Vision, and Machine Learning to assess sports performance through videos recorded on a smartphone. It analyzes factors such as movement, speed, posture, and technique to generate performance scores and personalized feedback. The platform helps identify and connect talented athletes, especially those from rural and underserved areas, with coaches and sports organizations
4	HERITAGE AND CULTURE	Student Innovation	SOFTWARE	Ideas that showcase the rich cultural heritage and traditions of India
5	MED TECH / BIOTECH / HEALTH TECH	Health Data Information & Management System Mobile Application (HDIMS)	SOFTWARE	Hospitals/Departments shall be able to enter and Update Data in the Mobile Application which can be viewed by the Super Admin dynamically for efficient Implementation of Health and Family Welfare Schemes, other Health programmes and provide key inputs for policy formulation and appropriate programme interventions. This Mobile Application will facilitates the flow of physical performance from the Facility level to the Sub-district, District and State/Union Territory level using Health Data Information & Management System Mobile Application (HDIMS) interface
6	MED TECH / BIOTECH / HEALTH TECH	Development of portable device (non-contact device) for measurement of eye pressure in glaucoma patients for usage at home.	HARDWARE	Background: Glaucoma is a leading cause of blindness worldwide, and managing intraocular pressure (IOP) is crucial for preventing disease progression. Description: Traditional methods of measuring IOP, such as Goldmann applanation tonometry, require direct contact with the eye, topical anaesthesia, and skilled personnel, making them less accessible for regular monitoring. Non-contact methods, such as air-puff tonometry, offer a more convenient alternative but are often limited to clinical settings due to their size and cost. Expected Solution: Problem statement is to develop a cost-effective, portable, and non-contact device that can measure IOP accurately and reliably with safety. This device should be easy to use, and allowing patients to use at home.
7	AGRICULTURE, FOOD TECH & RURAL DEVELOPMENT	AI-Driven Crop Disease Prediction and Management System	SOFTWARE	Background: Crop diseases can devastate yields, leading to significant financial losses for farmers. Early detection and timely intervention are crucial for effective management. Description: Develop an AI-driven system that analyzes crop images and environmental data to predict potential disease outbreaks. This system will provide farmers with actionable insights and treatment recommendations to mitigate risks. Expected Solution: A mobile and web-based application that utilizes machine learning algorithms to identify crop diseases and suggest preventive measures and treatments based on real-time data.
8	AGRICULTURE, FOOD TECH & RURAL DEVELOPMENT	Smart Irrigation System for Precision Farming	HARDWARE	Background: Water scarcity is a major challenge in agriculture, leading to inefficient water use and reduced crop yields. Traditional irrigation methods often result in overuse or underuse of water. Description: Develop a smart irrigation system that utilizes sensors and IoT technology to monitor soil moisture levels and weather conditions. This system will provide real-time data to farmers, enabling them to make informed decisions on irrigation scheduling, thus optimizing water usage and enhancing crop yield. Expected Solution: A smart irrigation module integrated with moisture sensors and a mobile application to provide farmers with timely irrigation alerts and recommendations based on real-time data.
9	SMART VEHICLES	Automated Bus Scheduling and Route Management System for Delhi Transport Corporation	SOFTWARE	Efficient bus scheduling and route planning are essential for reliable public transportation. Currently, DTC relies heavily on manual scheduling and route-planning methods, which are time-consuming and prone to errors. The proposed Automated Bus Scheduling and Route Management System uses algorithms, data analytics, and GIS technology to automate bus and crew scheduling. It supports both linked duty scheduling, where a crew member remains with the same bus throughout the duty, and unlinked duty scheduling, where buses can be handed over to other crew members after completing trips and required rest periods. The system also maps existing bus routes, allows users to draw proposed new routes, and automatically highlights overlapping sections. The main objective is to optimize bus and crew utilization, improve route coverage, reduce scheduling conflicts, and enhance service reliability. The final output will be a user-friendly prototype with scheduling, route mapping, overlap detection, dashboards, and reporting features for DTC planners and managers.
10	SMART VEHICLES	Student Innovation	HARDWARE	Creating intelligent devices to improve the commutation sector
11	TRANSPORTATION & LOGISTICS	Smart College Traffic & Parking Management System	SOFTWARE/HARDWARE	During college opening/closing times and events, many vehicles enter and leave the campus at the same time. This causes traffic congestion, parking problems, and difficulty finding available parking spaces. Develop a software/hardware system that monitors vehicle entry/exit and available parking spaces using sensors or cameras. The system should show available parking areas, guide vehicles to free spaces, and provide traffic information to the campus security/administration through a dashboard.
12	ROBOTICS & DRONES	Drone–Ground Robot Cooperative System for Disaster Debris Mapping	SOFTWARE/HARDWARE	After earthquakes, cyclones, landslides, or building collapses, rescue teams may not know which areas are safe to enter. A drone can first survey and map the disaster area, identifying debris, blocked paths, hazardous regions, and possible points of interest. A ground robot can then be automatically or remotely guided to selected locations for detailed inspection using cameras and sensors. The system should combine drone mapping, computer vision, robot navigation, sensor data, and a centralized rescue dashboard to provide an updated map of safe, risky, inaccessible, and priority areas. A software/hardware system is needed to support cooperative drone and ground-robot operations for disaster assessment.
13	ROBOTICS & DRONES	Drone-Based Campus/Building Inspection System	SOFTWARE/HARDWARE	Checking large buildings, rooftops, solar panels, towers, or other difficult-to-reach areas manually can be time-consuming and sometimes unsafe. Develop a software/hardware drone system that captures images or videos of selected areas and uses computer vision to identify visible problems such as cracks, damaged surfaces, or obstacles. The system should generate an inspection report.

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14	CLEAN & GREEN TECHNOLOGY	Smart Waste Segregation System	SOFTWARE/HARDWARE	In many places, wet waste and dry/recyclable waste are mixed together , making waste processing and recycling difficult. Develop a software/hardware system that uses sensors and/or a camera with AI image classification to identify waste and help separate it into categories such as wet waste, plastic, paper, and metal.
15	CLEAN & GREEN TECHNOLOGY	Smart Water Leakage & Wastage Detection System	SOFTWARE/HARDWARE	Water leaks in college buildings, hostels, homes, and public places may continue for a long time without being noticed, resulting in significant water wastage. Develop a software/hardware system that monitors water flow and usage using sensors. The system should detect unusual water flow, identify possible leakage, send alerts, and display water consumption through a dashboard.
16	TOURSIM	Tourist Crowd Management	SOFTWARE	Popular tourist destinations often experience heavy crowds during peak hours and seasons, resulting in long waiting times, traffic congestion, overcrowding, and reduced visitor satisfaction. At the same time, nearby attractions may remain less crowded and underused. Tourists and tourism authorities often lack a clear way to understand crowd conditions and their impact on different tourist locations, making effective crowd management difficult.
17	RENEWABLE / SUSTAINABLE ENERGY	Smart Biomass Moisture Detection System	SOFTWARE/HARDWARE	Agricultural wastes such as rice husk, straw, coconut shells, and wood residues are commonly used as biomass for renewable energy production. However, different moisture levels can affect their burning efficiency, energy output, and storage quality. Excess moisture can reduce energy production and may also cause fungal growth or spoilage during storage. A simple system is needed to monitor the moisture and temperature of biomass materials and determine whether they are suitable for energy production.
18	RENEWABLE / SUSTAINABLE ENERGY	Smart Solar Panel Shade Detection System	SOFTWARE/HARDWARE	Solar panels can experience a significant reduction in power generation when temporary or permanent shadows from trees, buildings, vehicles, or nearby objects fall on their surface. Users may not easily identify whether a decrease in power output is caused by shading or other factors. This can lead to reduced energy generation and inefficient use of the solar installation. A system is needed to monitor the solar panel's light conditions and power output and identify unusual reductions in generation caused by shading.
19	BLOCKCHAIN & CYBER SECURITY	Secure Digital Consent Tracking System	SOFTWARE	Users regularly provide consent to websites and applications to collect, store, and use their personal information, but they often cannot easily remember what information they agreed to share, when they gave consent, or how long the permission remains valid. Consent records may also be changed or become difficult to verify later. A software-based system is needed to maintain a secure and transparent history of user consent, allowing users to view, manage, and withdraw their permissions while providing organizations with a reliable record of consent.
20	Smart Education	Learning Resource Quality and Consistency Checker	SOFTWARE	Students use textbooks, websites, videos, and AI-generated materials from multiple sources, which may contain outdated, incorrect, duplicated, or conflicting information. A software-based system is needed to compare educational resources, identify inconsistencies and outdated content, and provide a reliability report, helping students and teachers choose accurate and suitable learning materials.
21	TOYS & GAMES	Preservation and Engagement of Traditional Indian Games	SOFTWARE	Many traditional Indian games that were once commonly played by children and communities are becoming less familiar to younger generations. Changes in lifestyle, limited awareness of regional variations, and the increasing preference for modern forms of entertainment have contributed to the gradual decline in participation and knowledge of these games. In many cases, information about the rules, cultural background, variations and traditional methods of playing such games is scattered or passed down informally. There is a need to address the challenge of making traditional Indian games more accessible and engaging to younger generations while preserving the cultural knowledge associated with them.
22	TOYS & GAMES	Gamified Programming Learning Platform	SOFTWARE	Beginners often find programming difficult because learning mainly involves understanding syntax, solving repetitive coding exercises, and identifying errors without an engaging learning environment. This can make it difficult for learners to develop logical thinking and problem-solving skills while maintaining interest in programming. There is a need to address the challenge of making programming concepts more
23	DISASTER MANAGEMENT	Emergency Resource Allocation System	SOFTWARE	During a disaster, rescue teams, ambulances, shelters, and essential supplies may not be available where they are needed most. It can be difficult for authorities to decide what should be sent first and where. Develop a solution that can help manage and distribute emergency resources effectively.
24	SPACE AND TECHNOLOGY	Making Space Science and Mission Operations Understandable to Students	SOFTWARE	Space missions involve several complex concepts, including launch sequences, orbital motion, satellite operations, communication, payloads, mission stages and spacecraft status. These concepts can be difficult for school and college students to understand because they are generally presented through highly technical material or theoretical explanations. Students often have limited opportunities to interact with or explore these concepts in an intuitive manner. There is a need to address the challenge of improving students' understanding of fundamental space science and mission operations in a way that makes complex concepts easier to explore and comprehend.
25	MISCELLANEOUS	Student Innovation	SOFTWARE/HARDWARE	Identify a meaningful real-world problem and develop an innovative, feasible technology-based solution to address it.